Session 9

Q. 1) Create a SQL script to insert a new order into an Orders table, then use COMMIT to save the transaction and verify that the new order persists after reconnectiong to the database.

Ans. insert into orders

(user_id, transaction)

values

(6, 10000);

start transaction;

update orders set transaction = transaction - 5000 where order_id = 13;

commit;

Q. 2) Simulate a Zomato-style food order process: insert two new items into an OrderItems table, then use ROLLBACK to undo the changes before commiting. Check that no new items remain in the table after rollback.

Ans. start transaction;

insert into orderItems

(item_name, payment)

values

('burger', 100),

('sandwich', 200);

rollback;

Q. 3) In a Flipkart-like shopping cart scenario, use SAVEPOINT to make a point after adding a product to the Cart table, add another product, then use ROLLBACK TO SAVEPOINT to undo only the last addition. Show the final contents of the Cart table.

Ans. start transaction;

```
insert into cart
(product_name, price)
values
('samsung mobile', 50000),
('mobile charger', 3500);

savepoint sp1;
insert into cart
(product_name, price)
values
('polo tshirt', 2500);

rollback to sp1;
commit;
```

Q. 4) Write s trigger on a Wallet table that automatically deducts the purchase amount from the user's balance whenever a new transaction is inserted, similar to how Paytm updates wallet balance after a payment.

Ans. delimiter $$

```
create trigger deduct_amount
after insert on payments
for each row
begin
update wallet
```

```
set total_amount = total_amount - new.amount

where user_id = new.user_id;

end $$

delimiter ;


insert into payments

(amount, payment_date)

values

-- (500, '2026-11-26');

(10000, '2026-12-25');
```

Q. 5) use chatgpt to generate sql test scenario where a trigger incorrectly updates a user's balance after an expense entry. paste the scenario and your corrected trigger code, explaining how you fixed the error.Hint: Ask chatgpt for a buggy example and a test case that exposes the bug.

Ans.

Suppose we have a personal expense tracking system with two tables:

- users – stores user details and their account balance.
- expenses – stores expenses entered by each user.

Whenever a new expense is added, a trigger should automatically subtract the expense amount from the correct user's balance.

Buggy trigger code:

```
DELIMITER $$

CREATE TRIGGER buggy_expense_trigger
AFTER INSERT ON expenses
FOR EACH ROW
BEGIN
   UPDATE users
   SET balance = balance - NEW.amount;
END $$

DELIMITER ;
```

Corrected trigger code:

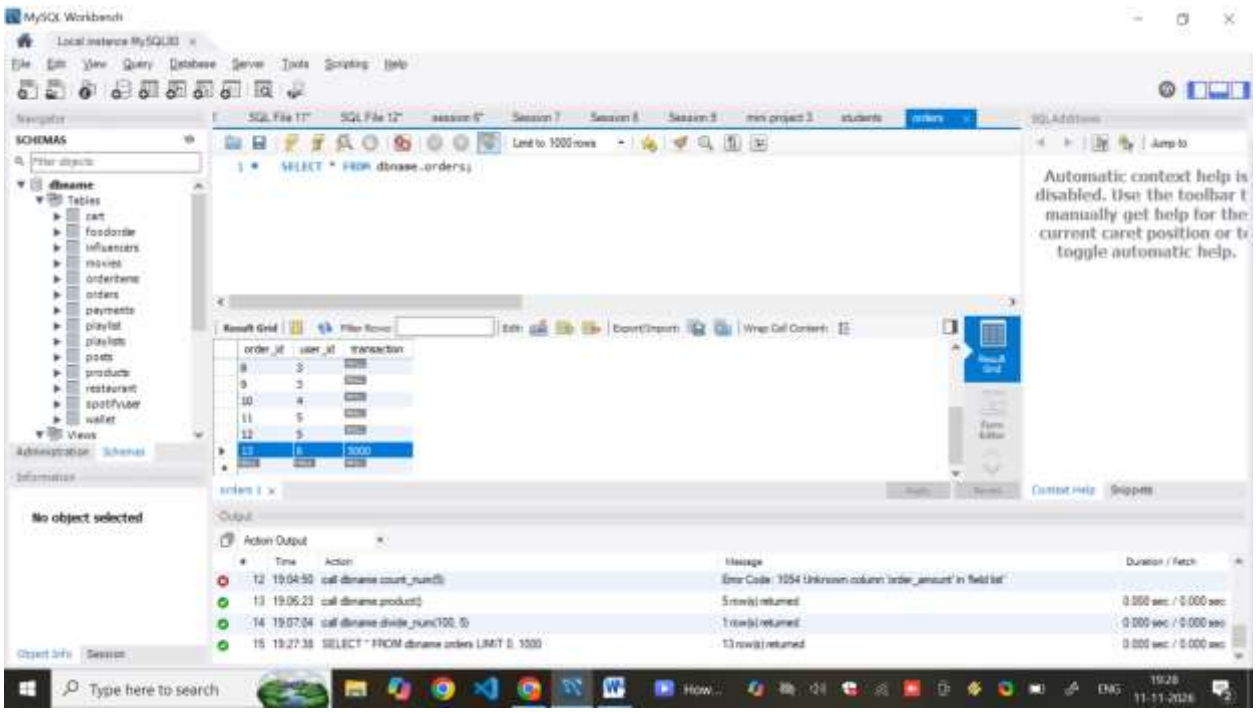
```
DELIMITER $$

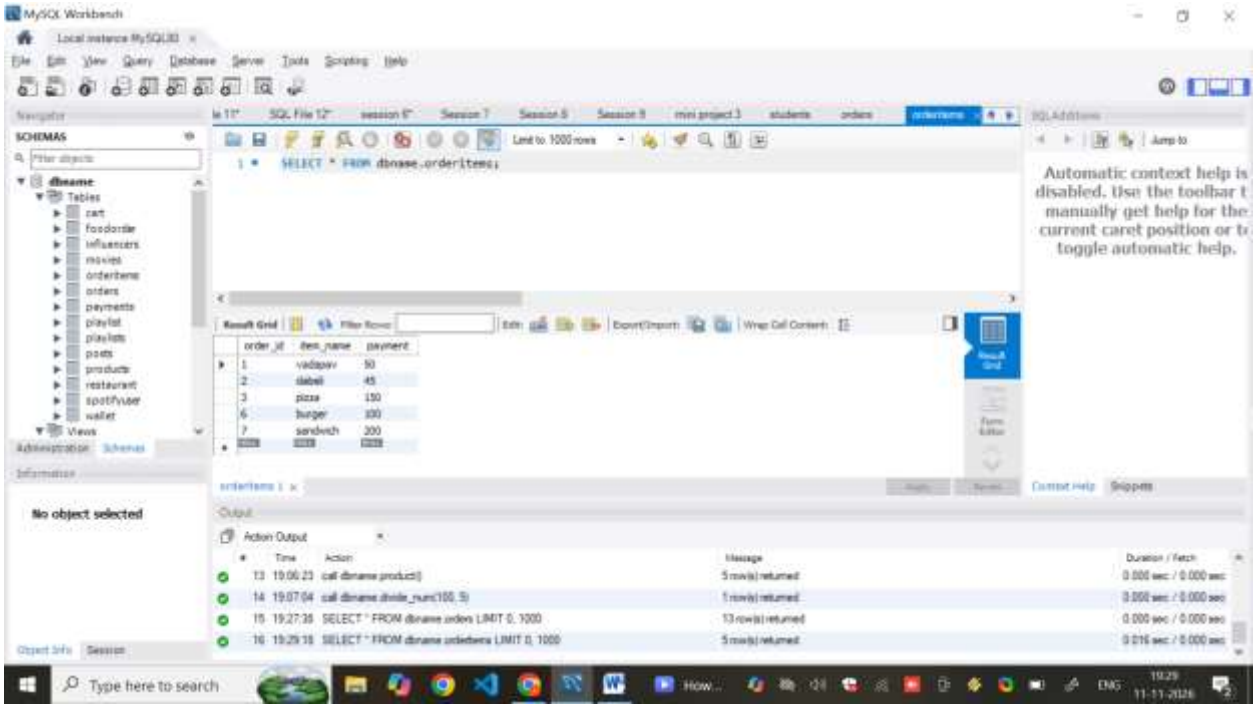
CREATE TRIGGER correct_expense_trigger
AFTER INSERT ON expenses
FOR EACH ROW
BEGIN
   UPDATE users
   SET balance = balance - NEW.amount
   WHERE user_id = NEW.user_id;
END $$

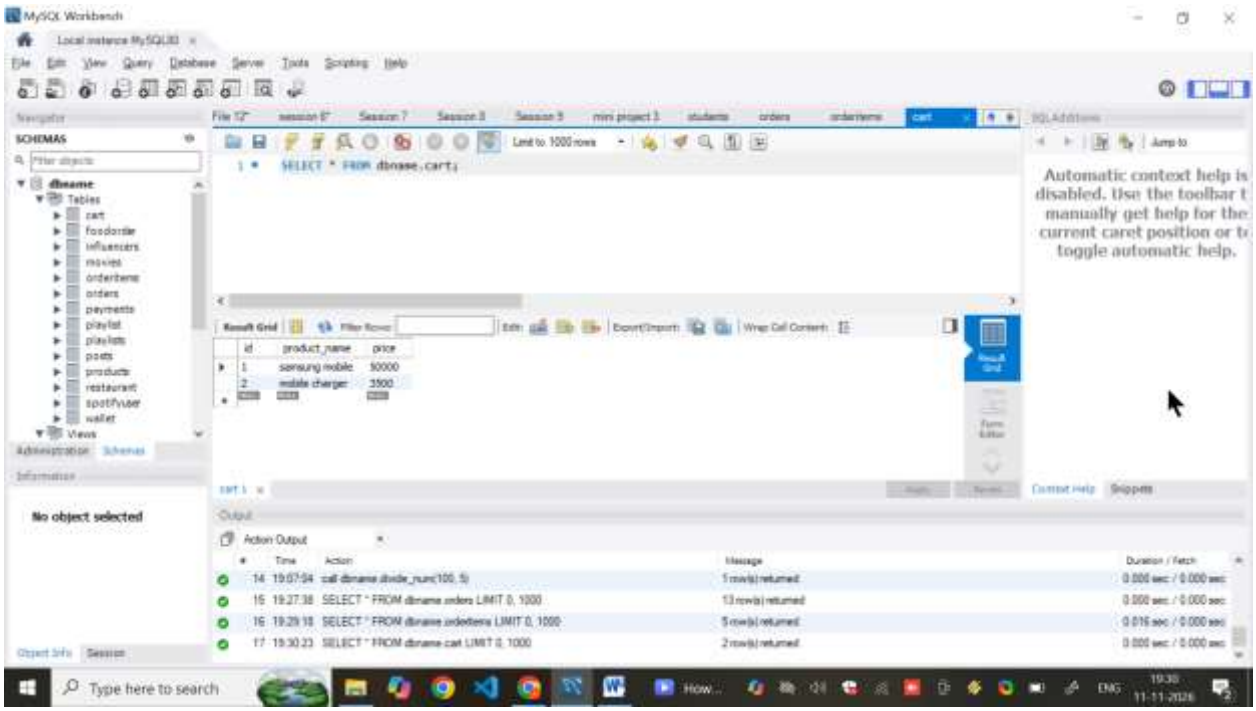
DELIMITER ;
```

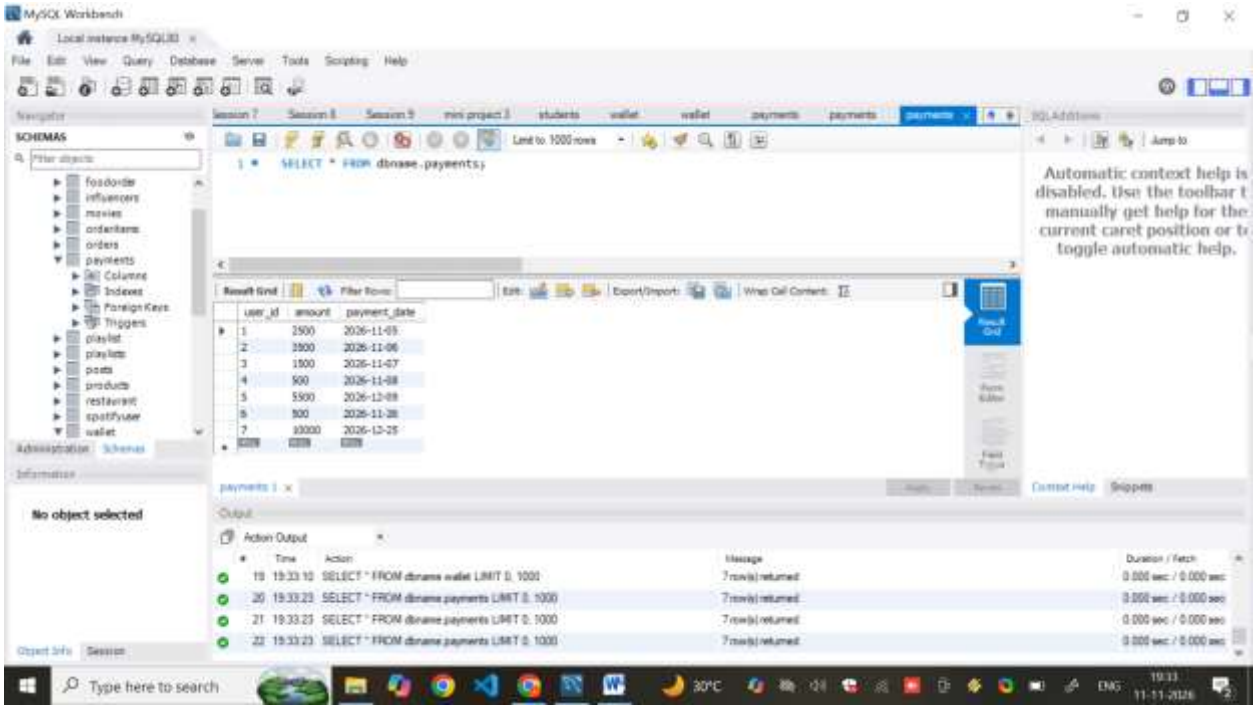
Explanation:

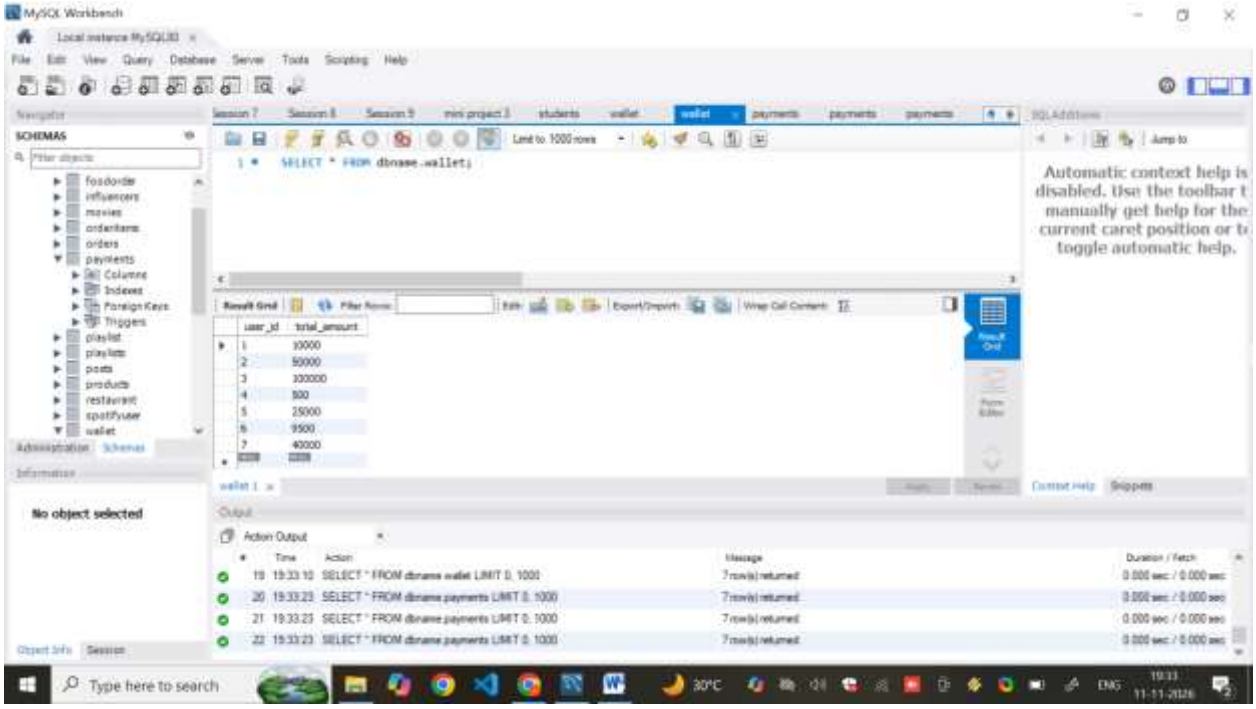
1. I identified that the buggy trigger was updating all users' balances.
2. The problem was the missing `WHERE` condition.
3. I added where user_id = new.user_id to update only the user who entered the expense.
4. new.amount gets the newly entered expense amount.
5. new.user_id identifies the user who made the expense.
6. I tested the corrected trigger and confirmed that only the correct user's balance was reduced.




MySQL Workbench
SELECT * FROM dbname.orderitems;
No object selected
Type here to search


MySQL Workbench
SELECT * FROM dbname.cart;
No object selected
Type here to search


MySQL Workbench
SELECT * FROM dbnase.payments;
user_id amount payment_date
1 1500 2026-11-05
2 3500 2026-11-06
3 1500 2026-11-07
4 500 2026-11-08
5 5500 2026-12-09
6 500 2026-11-28
7 10000 2026-12-25
No object selected
Automatic context help is disabled. Use the toolbar t manually get help for the current caret position or t toggle automatic help.
Type here to search


MySQL Workbench
SELECT * FROM dbnase.wallet;
user_id total_amount
1 10000
2 50000
3 100000
4 500
5 25000
6 9500
7 40000
No object selected
Automatic context help is disabled. Use the toolbar t manually get help for the current caret position or t toggle automatic help.
Type here to search